

Relational Algebra Final Project

UFC Database
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CHAMPIONS FROM PAST THREE YEARS $\leftarrow \pi_{\text{Champions.Champion}, \text{Champions.WeightClass}}$
($\sigma_{\text{SCHEDULE.DateOfMatch} \geq \text{Today}() - 1095}$ (CHAMPIONS $\bowtie_{\text{CHAMPION.MatchName} = \text{SCHEDULE.MatchName}}$
SCHEDULE))

SUMMARY OF FIGHT STYLES "Fight Style", "Number Of TakeDowns", "Number Of Wins" $\leftarrow \text{FIGHTER.FightStyle}$ $\bowtie$
COUNT(FIGHTER.#OfWins), COUNT(MATCHES.TakeDown) (FIGHTER $\bowtie_{\text{FIGHTER.PlayerID} = \text{MATCHES.Player1ID OR}$
FIGHTER.PlayerID = MATCH.Player2ID MATCHES))

ACTIVE FIGHTERS $\leftarrow \pi_{\text{FIGHTER.FirstName}, \text{FIGHTER.Retired?}, \text{FIGHTER.Country},$
FIGHTER.FightStyle ($\sigma_{\text{FIGHTER.Retired?} = \text{'no'}}$ (FIGHTER))

MATCHES FROM 2023 $\leftarrow \pi_{\text{MATCHES.MatchName}, \text{MATCHES. \#RoundsFought}, \text{MATCHES.TakeDown},$
SCHEDULE.DateOfMatch ($\sigma_{\text{SCHEDULE.DateOfMatch} \geq 01/01/2023 \text{ AND } \text{SCHEDULE.DateOfMatch} < 01/01/2024}$ (MATCHES $\bowtie$
MATCHES.MatchName = SCHEDULE.MatchName SCHEDULE))

FIGHTERS WHO MISSED THEIR WEIGHT CLASS $\leftarrow \pi_{\text{FIGHTER.FirstName}, \text{FIGHTER.LastName},$
FIGHTER.WeightMissed, FIGHTER.DateOfWeighIn, FIGHTER.Fine ($\sigma_{\text{FIGHTER.WeightMissed} \geq 10}$ (FIGHTER))

FUTURE MATCHES $\leftarrow \pi_{\text{SCHEDULE.DateOfMatch}, \text{SCHEDULE.TimeOfMatch}, \text{MATCHES.Player1FirstName},$
MATCHES.Player1LastName, MATCHES.Player2FirstName, MATCHES.Player2LastName. SCHEDULE.Location ($\sigma_{\text{DateOfMatch} \leq \text{Today}()}$
(SCHEDULE $\bowtie_{\text{SCHEDULE.MatchName} = \text{MATCHES.MatchName}}$ MATCHES))